

MTH-RA, Real Analysis

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General Information About the Course

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The classical motivation for measure theory goes back more than two thousand years: the Greeks wanted to measure the area of a circle, the volume of a ball, and other regions in the plane and in space. This eventually evolved into an abstract theory: instead of working only in $\mathbb{R}^d$, we work with an arbitrary set Ω and ask how to assign a consistent notion of “size” to certain of its subsets.

A key obstruction appears immediately:

- It is **not possible**, in general, to consistently measure *every* subset of a set (even of $\mathbb{R}^d$).
- Instead, we single out a suitable collection of “good” subsets (we call them the **measurable sets**) and only attempt to measure those.

This leads to the first fundamental definition of the course: the notion of a σ -algebra.

Definition 1.1 (σ -algebra). Let Ω be a set. A collection Σ of subsets of Ω is called a σ -algebra (or σ -field) on Ω if:

- (i) $\Sigma \neq \emptyset$ (equivalently, $\Omega \in \Sigma$);
- (ii) (closed under complements) if $A \in \Sigma$, then $A^c := \Omega \setminus A \in \Sigma$;
- (iii) (closed under countable unions) if $A_1, A_2, A_3, \dots \in \Sigma$, then $\bigcup_{n=1}^{\infty} A_n \in \Sigma$.

The pair (Ω, Σ) is called a **measurable space**, and the sets $A \in \Sigma$ are called **measurable sets**.

Remark 1.1. Condition (i) simply rules out the degenerate empty collection, which would vacuously satisfy (ii) and (iii) but is of no interest. Some textbooks instead require $\emptyset \in \Sigma$ directly; this is equivalent given (ii)–(iii) and a nonempty Σ .

Proposition 1.1

Every σ -algebra Σ on Ω satisfies:

- (a) $\emptyset \in \Sigma$ and $\Omega \in \Sigma$;
- (b) Σ is closed under countable intersections: if $A_1, A_2, \dots \in \Sigma$ then $\bigcap_{n=1}^{\infty} A_n \in \Sigma$;
- (c) Σ is closed under finite unions and finite intersections (in particular under set differences).

Proof. (a) Since $\Sigma \neq \emptyset$, pick $A \in \Sigma$. By (ii), $A^c \in \Sigma$, and by (iii) (a union of the two, or of A and infinitely many copies of it), $A \cup A^c = \Omega \in \Sigma$. Applying (ii) again, $\Omega^c = \emptyset \in \Sigma$.

(b) By De Morgan's law,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c. \quad (1.1)$$

Each $A_n^c \in \Sigma$ by (ii), the union is in Σ by (iii), and its complement is in Σ by (ii) again.

(c) A finite union $A_1 \cup \dots \cup A_k$ is a countable union with $A_n = \emptyset$ for $n > k$ (using $\emptyset \in \Sigma$ from (a)), so it lies in Σ by (iii). Finite intersections follow similarly from (b), and $A \setminus B = A \cap B^c \in \Sigma$. ■

Remark 1.2. The definition does *not* require the sets A_n in condition (iii) to be pairwise disjoint; the countable union of *any* sequence of measurable sets is measurable.

Example 1.1 (Trivial and power set σ -algebras)

For any Ω :

- $\{\emptyset, \Omega\}$ is the smallest possible σ -algebra on Ω .
- The power set $\mathcal{P}(\Omega)$ (all subsets of Ω) is the largest possible σ -algebra on Ω ; if $|\Omega| = n < \infty$ then $|\mathcal{P}(\Omega)| = 2^n$.

Example 1.2 (Generated σ -algebra)

Let X be an arbitrary collection of subsets of Ω . There is a **smallest** σ -algebra $\sigma(X)$ containing X , called the σ -algebra **generated by** X . It can be constructed as

$$\sigma(X) = \bigcap \{ \Sigma : \Sigma \text{ is a } \sigma\text{-algebra on } \Omega, X \subseteq \Sigma \}, \quad (1.2)$$

i.e. the intersection of *all* σ -algebras containing X (an arbitrary intersection of σ -algebras is again a σ -algebra, and $\mathcal{P}(\Omega)$ is one such σ -algebra containing X , so this intersection is nonempty and well defined).

Lemma 1.2

Let Ω be a set and $\{\Sigma_\alpha\}_{\alpha \in J}$ an arbitrary collection of σ -algebras on Ω . Then $\bigcap_{\alpha \in J} \Sigma_\alpha$ is also a σ -algebra on Ω .

Proof. Every Σ_α contains $\emptyset$ and Ω , so $\Sigma := \bigcap_{\alpha \in J} \Sigma_\alpha$ is nonempty.

Suppose $A \in \Sigma$. Then $A \in \Sigma_\alpha$ for every α . Since Σ_α is a σ -algebra, it contains A^c , for every α . Therefore, $\Sigma := \bigcap_{\alpha \in J} \Sigma_\alpha$ also contains A^c .

Suppose $A_n \in \Sigma$ for $n \in \mathbb{N}$. Then $A_n \in \Sigma_\alpha$ for every α , for every n . Since Σ_α is a σ -algebra, it contains $\bigcup_{n \in \mathbb{N}} A_n$, for every α . Therefore, $\Sigma := \bigcap_{\alpha \in J} \Sigma_\alpha$ also contains $\bigcup_{n \in \mathbb{N}} A_n$. ■

Lemma 1.2 shows that $\sigma(X)$ from (1.2) is a σ -algebra.

Example 1.3 (A concrete generated σ -algebra)

Let $\Omega = \{1, 2, 3, 4\}$ and $X = \{\{1, 2\}, \{1, 4\}\}$. We compute $\sigma(X)$ by closing under complements and unions:

$$\{1, 2\}, \{3, 4\}, \{1, 4\}, \{2, 3\}, \{1, 2, 4\} \rightarrow \{1, 2\} \cup \{1, 4\}, \{3\}, \{2\}, \{1\}, \{4\}, \dots \quad (1.3)$$

Continuing this process eventually generates every subset of $\{1, 2, 3, 4\}$, so in this case $\sigma(X) = \mathcal{P}(\Omega)$, with 16 elements.

Example 1.4 (A proper subalgebra)

Let $\Omega = \{1, 2, 3, 4\}$ and $X = \{\{1, 2\}\}$. Closing under complement and union gives

$$\sigma(X) = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}, \quad (1.4)$$

which has only $4 = 2^2$ elements, which is a proper subset of $\mathcal{P}(\Omega)$ (which has 16 elements). This shows that a σ -algebra need not be the full power set.

Exercise 1.1 (Possible cardinalities). Let $\Omega = \{1, \dots, n\}$.

- (a) Show that every σ -algebra on a finite set Ω has cardinality equal to a power of 2.
- (b) For a fixed achievable cardinality 2^k , is the σ -algebra with that cardinality unique? Investigate how many distinct σ -algebras of cardinality 2^k exist on $\{1, \dots, n\}$.

Solution. Let Σ be a σ -algebra on Ω . We define a relation on Ω as follows:

$$x \sim y \iff \text{for every } A \in \Sigma, x \in A \text{ if and only if } y \in A. \quad (1.5)$$

This is clearly an equivalence relation. Hence, the equivalence classes partition Ω . Now, for a given $x \in \Omega$, let C_x be the equivalence class containing x . Then,

$$C_x = \bigcap_{A \in \Sigma, x \in A} A \in \Sigma, \quad (1.6)$$

as it's the intersection of sets in Σ . Furthermore, for $A \in \Sigma$,

$$A = \bigcup_{x \in A} C_x. \quad (1.7)$$

Hence, every element of Σ is the disjoint union of members of $\mathcal{C} = \{C_x : x \in \Omega\}$. Furthermore, **every** disjoint union of members of $\mathcal{C}$ is in Σ , since $\mathcal{C} \subseteq \Sigma$. Hence, the elements of Σ are in a bijection with the subsets of $\mathcal{C}$. Therefore,

$$|\Sigma| = 2^{|\mathcal{C}|}. \quad (1.8)$$

For the second part, $|\Sigma| = 2^k$ corresponds to a partition of Ω into k disjoint parts, which is the Stirling number of second kind, denoted as $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$. ■

Example 1.5 (Borel σ -algebra)

Let $\Omega = \mathbb{R}^d$ with its usual topology, and let X be the collection of all **open** subsets of $\mathbb{R}^d$. The generated σ -algebra

$$\mathcal{B}(\mathbb{R}^d) := \sigma(X) \quad (1.9)$$

is called the **Borel σ -algebra**. It contains every open set, every closed set (as complements of open sets), and all sets obtainable from these by countably many unions, intersections, and complements. In particular, on $\mathbb{R}$, it contains every interval (open, closed, or half-open).

It is useful to contrast a σ -algebra with a **topology**, since both are collections of subsets satisfying closure axioms.

Definition 1.2 (Topology). A **topology** τ on a set Ω is a collection of subsets of Ω (called **open sets**) such that:

- (i) $\emptyset, \Omega \in \tau$;
- (ii) τ is closed under **arbitrary** unions: if $\{O_j\}_{j \in J}$ is any family (not necessarily countable) of sets in τ , then $\bigcup_{j \in J} O_j \in \tau$;
- (iii) τ is closed under **finite** intersections: if $O_1, \dots, O_n \in \tau$, then $\bigcap_{i=1}^n O_i \in \tau$.

Remark 1.3 (Key differences). • A topology allows *arbitrary* (possibly uncountable) unions but only *finite* intersections.

- A σ -algebra allows only *countable* unions (and, as a consequence, countable intersections), but requires closure under **complements**, which a topology does not.

These are genuinely different axiom systems, even though both start from “a set together with a distinguished collection of subsets.”

Having fixed a measurable space (Ω, Σ) , we now assign a size to each measurable set.

Definition 1.3 (Measure). A **measure** on (Ω, Σ) is a function $\mu : \Sigma \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\} = [0, \infty]$ such that:

- (i) (countable additivity) if $A_1, A_2, A_3, \dots \in \Sigma$ are pairwise disjoint, then

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu(A_n); \quad (1.10)$$

(ii) μ is not identically $+\infty$ (i.e. $\mu(A) < \infty$ for at least one $A \in \Sigma$).

The triple (Ω, Σ, μ) is called a **measure space**.

Remark 1.4 (Motivation for additivity). If a measure is meant to behave like a physical quantity, e.g. volume of water in a bucket, then combining two *disjoint* quantities should simply add their sizes: 3 liters and 5 liters combined give 8 liters, with nothing created or destroyed. Countable additivity is the natural axiomatization of this idea.

Proposition 1.3 (The measure of the empty set)

Let μ be a measure on (Ω, Σ) satisfying countable additivity (condition (i) above), and suppose μ is not identically $+\infty$. Then $\mu(\emptyset) = 0$.

Proof. Apply additivity to the disjoint sequence $A, \emptyset, \emptyset, \dots$ for a fixed A with $\mu(A) < \infty$:

$$\mu(A) = \mu(A \cup \emptyset) = \mu(A) + \mu(\emptyset). \quad (1.11)$$

Since $\mu(A) < \infty$, we may subtract it from both sides to get $\mu(\emptyset) = 0$.

(If instead $\mu(\emptyset) = c > 0$ were allowed, one could show $\mu(\emptyset) = \mu(\emptyset \cup \emptyset) = 2c$, forcing $c \in \{0, \infty\}$; and if $\mu(\emptyset) = \infty$ then, applying additivity with A and $\emptyset$, $\mu(A) = \mu(A) + \infty = \infty$ for every A , i.e. μ is identically ∞ . This is exactly the degenerate case excluded by (ii).) ■

Remark 1.5. Some textbooks instead simply impose $\mu(\emptyset) = 0$ directly as part of the definition, in place of the “not identically ∞ ” condition; the two conventions agree on all non-degenerate measures. This mirrors the analogous convention issue for topologies (where $\emptyset, \Omega \in \tau$ is imposed directly).

Definition 1.4. Let (Ω, Σ, μ) be a measure space.

- μ is a **finite measure** if $\mu(\Omega) < \infty$.
- If μ is finite and $\mu(\Omega) = 1$, then μ is called a **probability measure**.
- μ is **σ -finite** if there exists a sequence $A_1, A_2, A_3, \dots \in \Sigma$ such that

$$\mu(A_n) < \infty \text{ for every } n, \quad \text{and} \quad \Omega = \bigcup_{n=1}^{\infty} A_n. \quad (1.12)$$

Remark 1.6. If $\mu(\Omega) < \infty$, one can always **normalize** μ by setting $\tilde{\mu}(A) = \mu(A)/\mu(\Omega)$ (assuming $\mu(\Omega) > 0$), producing a probability measure with the same measurable sets of measure zero.

Proposition 1.4

Let (M, Σ, μ) be a measure space.

(a) If $A_1 \subseteq A_2 \subseteq \cdots$ is an increasing sequence of measurable sets, then

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} \mu(A_n). \quad (1.13)$$

(b) If $A_1 \supseteq A_2 \supseteq \cdots$ is a decreasing sequence of measurable sets and $\mu(A_1) < \infty$, then

$$\mu \left(\bigcap_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} \mu(A_n). \quad (1.14)$$

Proof. (a) We define $B_1 = A_1$ and $B_n = A_n - A_{n-1}$ for $n \geq 2$. Then the B_n 's are pairwise disjoint, and $\bigcup_n B_n = \bigcup_n A_n$. As a result,

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \mu \left(\bigcup_{n=1}^{\infty} B_n \right) = \sum_{n=1}^{\infty} \mu(B_n) \quad (1.15)$$

$$= \lim_{N \rightarrow \infty} \sum_{n=1}^N \mu(B_n) = \lim_{N \rightarrow \infty} \mu \left(\bigcup_{n=1}^N B_n \right) \quad (1.16)$$

$$= \lim_{N \rightarrow \infty} \mu(A_N), \quad (1.17)$$

since $\bigcup_{n=1}^N B_n = A_N$.

(b) Define $B_n = A_1 - A_n$. Then B_n is an increasing sequence of measurable sets. Furthermore,

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A_1 - A_n) = A_1 - \bigcap_{n=1}^{\infty} A_n. \quad (1.18)$$

Claim 1.1: If A, B are measurable sets with $A \subseteq B$ and $\mu(A) < \infty$, then $\mu(B - A) = \mu(B) - \mu(A)$.

Proof. $B = A \cup (B - A)$ and this is a disjoint decomposition. Therefore,

$$\mu(B) = \mu(A) + \mu(B - A). \quad (1.19)$$

Rearranging gives the desired result. $\square$

By applying part (a), we get

$$\mu \left(\bigcup_{n=1}^{\infty} B_n \right) = \lim_{n \rightarrow \infty} \mu(B_n). \quad (1.20)$$

Applying [Claim 1.1](#) into both sides of the equation, we get

$$\mu(A_1) - \mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \mu(A_1) - \lim_{n \rightarrow \infty} \mu(A_n). \quad (1.21)$$

■

Remark 1.7. The condition that $\mu(A_1) < \infty$ (or at least $\mu(A_n) < \infty$ for some n) is important here. For example, consider

$$\Omega = \mathbb{N}, \quad \Sigma = \mathcal{P}(\mathbb{N}), \quad \mu = \text{counting measure}.$$

Here, $A_n = \{n, n+1, n+2, \dots\}$ is a decreasing sequence with $\mu(A_n) = \infty$ for every n . But

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \mu(\emptyset) = 0 \text{ and } \lim_{n \rightarrow \infty} \mu(A_n) = \infty. \quad (1.22)$$

Example 1.6

Lebesgue measure on $\mathbb{R}^d$ is not finite (the whole space has infinite volume), but it *is* σ -finite: taking $A_n = B(0, n)$, the ball of radius n centered at the origin, each A_n has finite volume and $\mathbb{R}^d = \bigcup_{n=1}^{\infty} A_n$.

Remark 1.8. The choice of Ω , Σ , and μ is entirely up to the application at hand. There is no canonical choice in general. However, certain combinations are used so pervasively that they become the default setting for the theory; the main example developed in this course will be $\mathbb{R}^d$ equipped with the Borel (or Lebesgue) σ -algebra and Lebesgue measure.

Definition 1.5 (Measurable function). Let (Ω, Σ) be a measurable space. A function $f : \Omega \rightarrow \mathbb{R}$ is called **measurable** (with respect to Σ) if, for every interval $I \subseteq \mathbb{R}$,

$$f^{-1}(I) := \{x \in \Omega : f(x) \in I\} \in \Sigma. \quad (1.23)$$

Remark 1.9. It suffices to check this condition for intervals of the special form $I = (a, \infty)$, $a \in \mathbb{R}$: since Σ is closed under complements, countable unions, and countable intersections, every interval (open, closed, half-open, bounded, or unbounded) can be built from intervals of this one type. Because,

$$\begin{aligned} (-\infty, a] &= (a, \infty)^c, \quad [a, \infty) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, \infty\right), \quad (-\infty, a) = [a, \infty)^c, \\ (a, b) &= (a, \infty) \cap (-\infty, b), \quad [a, b] = [a, \infty) \cap (-\infty, b], \\ [a, b) &= [a, \infty) \cap (-\infty, b), \quad (a, b] = (a, \infty) \cap (-\infty, b]; \end{aligned}$$

and preimage distributes union, intersections and complements. That's why many texts simply *define* measurability by requiring $f^{-1}((a, \infty)) \in \Sigma$ for all $a \in \mathbb{R}$.

Remark 1.10 (Measurability depends on the choice of Σ). Whether a given function is measurable depends entirely on the σ -algebra chosen on the domain. It is not an intrinsic property of the function alone.

Example 1.7

Let $\Omega = \mathbb{R}$ and $\Sigma = \{\emptyset, \mathbb{R}\}$ (the trivial σ -algebra).

- The constant function $f(x) = 1$ is measurable: for any interval I , $f^{-1}(I)$ is either $\mathbb{R}$ (if $1 \in I$) or $\emptyset$ (if $1 \notin I$), and both lie in Σ .
- The identity function $f(x) = x$ is **not** measurable: for a bounded interval such as $I = (0, 1)$, $f^{-1}(I) = (0, 1) \notin \{\emptyset, \mathbb{R}\}$.

This illustrates that measurability is relative to Σ : the same function can be measurable for one σ -algebra and not for another.

Example 1.8 (Basic examples of measurable functions)

The following are basic examples of measurable functions (with respect to the Borel σ -algebra, unless otherwise specified):

- Any constant function $f \equiv c$ is measurable.
- For a measurable set $A \subseteq \Omega$, the **characteristic function** of A is defined by

$$\chi_A(x) := \begin{cases} 1, & \text{if } x \in A; \\ 0, & \text{if } x \notin A. \end{cases} \quad (1.24)$$

This function is measurable: for $a \in \mathbb{R}$, the following sets are all measurable

$$\{x \in \Omega : \chi_A(x) > a\} = \begin{cases} \Omega, & a < 0; \\ A, & 0 \leq a < 1; \\ \emptyset, & a \geq 1. \end{cases} \quad (1.25)$$

- Every continuous function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is measurable with respect to the Borel σ -algebra: for any $a \in \mathbb{R}$, the set $\{x : f(x) > a\} = f^{-1}((a, \infty))$ is open (being the preimage of an open set under a continuous map), hence Borel measurable.
- Every monotone function $f : \mathbb{R} \rightarrow \mathbb{R}$ (increasing or decreasing) is measurable with respect to the Borel σ -algebra: for any $a \in \mathbb{R}$, the set $\{x : f(x) > a\}$ is always an interval (a ray, of one of the forms (c, ∞) , $[c, \infty)$, $(-\infty, c)$, or $(-\infty, c]$, depending on monotonicity and endpoint conventions), and every interval is Borel measurable.

Remark 1.11 (Measurable maps between measure spaces). More generally, if (Ω_1, Σ_1) and (Ω_2, Σ_2) are two measurable spaces, a function $f : \Omega_1 \rightarrow \Omega_2$ is called **measurable** if $f^{-1}(B) \in \Sigma_1$ for every $B \in \Sigma_2$. The definition above is the special case $\Omega_2 = \mathbb{R}$ with $\Sigma_2 = \mathcal{B}(\mathbb{R})$, the Borel σ -algebra.

It will be convenient for us to also allow our functions to take values in the extended real line $\overline{\mathbb{R}}$. The definition is the same as before: a function $f : \Omega \rightarrow \overline{\mathbb{R}}$ is **measurable** if the preimage of any extended interval is measurable, or, alternatively, if the sets $\{x \in \Omega : f(x) > a\}$ are measurable for each $a \in \mathbb{R}$. In this case, observe that the sets

$$\{x \in \Omega : f(x) = +\infty\} = \bigcap_{n=1}^{\infty} \{x \in \Omega : f(x) > n\}, \quad (1.26)$$

$$\{x \in \Omega : f(x) = -\infty\} = \left(\bigcup_{n=1}^{\infty} \{x \in \Omega : f(x) > -n\} \right)^c \quad (1.27)$$

are both measurable.

We shall adopt the convention that

$$0 \cdot (\pm\infty) = 0 \text{ and } (\pm\infty) + \pm(\mp\infty) = 0. \quad (1.28)$$

Let $\mathcal{M} = \mathcal{M}(\Omega, \Sigma)$ denote the set of measurable functions $f : \Omega \rightarrow \mathbb{R}$ (or $\overline{\mathbb{R}}$).

Proposition 1.5

Let $f, g \in \mathcal{M}$ and $c \in \mathbb{R}$. Then each of the following functions is also measurable:

- (a) cf ;
- (b) $f \pm g$;
- (c) fg ;
- (d) $\frac{f}{g}$, if $g \neq 0$ everywhere;
- (e) $|f|$.

Proof. We are going to use the trick that f is measurable if the sets $\{x \in \Omega : f(x) > a\}$ are all measurable.

- (a) If $c = 0$, then $cf = 0$, which is a constant function, so it's measurable. For $c > 0$,

$$\{x \in \Omega : c \cdot f(x) > a\} = \left\{x \in \Omega : f(x) > \frac{a}{c}\right\} \quad (1.29)$$

is measurable since f is a measurable function. For $c < 0$,

$$\{x \in \Omega : c \cdot f(x) > a\} = \left\{x \in \Omega : f(x) < \frac{a}{c}\right\} \quad (1.30)$$

is measurable since f is a measurable function.

(b) Note that (assuming $f(x), g(x)$ are finite)

$$\begin{aligned} f(x) + g(x) > a &\iff f(x) > a - g(x) \\ &\iff \exists q \in \mathbb{Q} \text{ such that } f(x) > q > a - g(x) \\ &\iff \exists q \in \mathbb{Q} \text{ such that } f(x) > q \text{ and } g(x) > a - q. \end{aligned}$$

$$\begin{aligned} &\{x \in \Omega : f(x) + g(x) > a\} \\ &= \{x \in \Omega : \text{both } f(x) \text{ and } g(x) \text{ are finite, and } f(x) + g(x) > a\} \\ &\quad \cup (\{x \in \Omega : f(x) = \infty\} \cap \{x \in \Omega : g(x) > -\infty\}) \\ &\quad \cup (\{x \in \Omega : g(x) = \infty\} \cap \{x \in \Omega : f(x) > -\infty\}) \\ &\quad \cup X, \end{aligned} \tag{1.31}$$

where X is

$$X = \begin{cases} \{x \in \Omega : f(x) = +\infty, g(x) = -\infty\} \cup \{x \in \Omega : g(x) = +\infty, f(x) = -\infty\}, & a < 0, \\ \emptyset, & a \geq 0. \end{cases} \tag{1.32}$$

Therefore,

$$\begin{aligned} &\{x \in \Omega : f(x) + g(x) > a\} \\ &= \bigcup_{q \in \mathbb{Q}} (\{x \in \Omega : f(x) > q\} \cap \{x \in \Omega : g(x) > a - q\} \cap \{x \in \Omega : f(x), g(x) \in \mathbb{R}\}) \\ &\quad \cup (\{x \in \Omega : f(x) = \infty\} \cap \{x \in \Omega : g(x) > -\infty\}) \\ &\quad \cup (\{x \in \Omega : g(x) = \infty\} \cap \{x \in \Omega : f(x) > -\infty\}) \\ &\quad \cup X. \end{aligned} \tag{1.33}$$

This is a measurable set since it's made up of measurable sets by taking union and intersection.

(c) We're gonna prove an intermediate lemma first.

Claim 1.2: If f is measurable, then so is f^2 .

Proof. If $a < 0$, then $\{x \in \Omega : f(x)^2 > a\} = \Omega$, is measurable. For $a \geq 0$,

$$\{x \in \Omega : f(x)^2 > a\} = \{x \in \Omega : f(x) > \sqrt{a}\} \cup \{x \in \Omega : f(x) < -\sqrt{a}\} \tag{1.34}$$

is measurable. □

Now notice that

$$fg = \frac{1}{4} [(f+g)^2 - (f-g)^2], \tag{1.35}$$

is indeed measurable.

(d) It suffices to show that $\frac{1}{g}$ is measurable.

$$\left\{x \in \Omega : \frac{1}{g(x)} > a\right\} = \begin{cases} \{x \in \Omega : 0 < g(x) < \frac{1}{a}\}, & a > 0, \\ \{x \in \Omega : g(x) > 0\}, & a = 0, \\ \{x \in \Omega : \frac{1}{a} < g(x) < 0\}, & a < 0. \end{cases} \quad (1.36)$$

These are all measurable since g is measurable.

(e) If $a < 0$, then $\{x \in \Omega : |f(x)| > a\} = \Omega$, is measurable. For $a \geq 0$,

$$\{x \in \Omega : |f(x)| > a\} = \{x \in \Omega : f(x) > a\} \cup \{x \in \Omega : f(x) < -a\} \quad (1.37)$$

is measurable. ■

Proposition 1.6

Let $(f_n)_{n=1}^{\infty}$ be a countable sequence of measurable functions $\Omega \rightarrow \overline{\mathbb{R}}$. Then each of the following functions is measurable:

- (a) $f(x) = \sup_n f_n(x)$;
- (b) $f(x) = \inf_n f_n(x)$;
- (c) $f(x) = \limsup_{n \rightarrow \infty} f_n(x)$;
- (d) $f(x) = \liminf_{n \rightarrow \infty} f_n(x)$.

Consequently, if $f_n \rightarrow f$ pointwise, then f is measurable.

Proof. (a) Note that

$$\sup_n f_n(x) > a \iff f_n(x) > a \text{ for some } n. \quad (1.38)$$

Therefore,

$$\left\{x \in \Omega : \sup_n f_n(x) > a\right\} = \bigcup_n \{x \in \Omega : f_n(x) > a\}, \quad (1.39)$$

is measurable.

(b) Note that

$$\inf_n f_n(x) = - \left(\sup_n (-f_n(x)) \right). \quad (1.40)$$

(c) Follows from the fact that

$$\limsup_{n \rightarrow \infty} f_n(x) = \inf_m \sup_{n \geq m} f_n(x). \quad (1.41)$$

(d) Follows from the fact that

$$\liminf_{n \rightarrow \infty} f_n(x) = \sup_m \inf_{n \geq m} f_n(x). \quad (1.42)$$

When $f_n \rightarrow f$ pointwise, then the limit, limsup and liminf coincides. ■

Remark 1.12. In the case of complex-valued functions, we say that $f : \Omega \rightarrow \mathbb{C}$ is measurable if its real part and imaginary parts are both measurable.

Definition 1.6 (Almost everywhere). Let (Ω, Σ, μ) be a measure space. We say that a property (P) holds μ -**almost everywhere** (or simply **almost everywhere**, when the measure μ is clear from context), abbreviated μ -a.e. or a.e., if there exists a set $N \in \Sigma$ with $\mu(N) = 0$ such that (P) holds on N^c . Such a set N is called a **null set**.

Example 1.9

Typical almost-everywhere statements include:

- $f = g$ almost everywhere, meaning $\{x \in \Omega : f(x) \neq g(x)\}$ is contained in a null set;
- $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ almost everywhere, meaning the set of x for which this limit fails to hold is contained in a null set.

Remark 1.13 (Subsets of null sets need not be measurable). If $\mu(N) = 0$ and $B \subseteq N$, one would like to conclude that $\mu(B) = 0$ as well. However, this conclusion presupposes $B \in \Sigma$, which need not be the case, since Σ may fail to contain every subset of a null set.

Definition 1.7 (Complete measure). A measure μ on (Ω, Σ) is called **complete** if its domain Σ contains every subset of every μ -null set; that is, whenever $N \in \Sigma$ satisfies $\mu(N) = 0$ and $B \subseteq N$, then $B \in \Sigma$ (and consequently $\mu(B) = 0$).

Remark 1.14. Every measure can be made complete by slightly enlarging its σ -algebra, as follows. Let (Ω, Σ, μ) be a measure space. Let

$$\mathcal{N} := \{N \in \Sigma : \mu(N) = 0\}, \quad \bar{\Sigma} := \{A \cup B : A \in \Sigma \text{ and } B \subseteq N \text{ for some } N \in \mathcal{N}\}. \quad (1.43)$$

$\bar{\Sigma}$ is a σ -algebra. Indeed:

1. Take $A \cup B \in \bar{\Sigma}$, where $A \in \Sigma$ and $B \subseteq N$ for some $N \in \mathcal{N}$. Then

$$(A \cup B)^c = A^c \cap B^c = A^c - (B \cap A^c). \quad (1.44)$$

We're gonna use $W - U = (W - V) \cup (V - U)$ for $U \subseteq V \subseteq W$, with $U = B \cap A^c$, $V = N \cap A^c$, $W = A^c$. Then the RHS of (1.44) becomes,

$$[A^c - (N \cap A^c)] \cup [(N \cap A^c) - (B \cap A^c)]. \quad (1.45)$$

$A^c - (N \cap A^c) \in \Sigma$ since it's made up of elements of Σ using complementation and finite intersection. $(N \cap A^c) - (B \cap A^c) \subseteq N$. Therefore, $(A \cup B)^c \in \bar{\Sigma}$.

2. Let $A_i \cup B_i \in \bar{\Sigma}$ be a countable sequence of sets, with $A_i \in \Sigma$ and $B_i \subseteq N_i$ with $\mu(N_i) = 0$. Then

$$\mu\left(\bigcup_{i=1}^{\infty} N_i\right) \leq \sum_{i=1}^{\infty} \mu(N_i) = 0, \quad (1.46)$$

implying that $\bigcup_{i=1}^{\infty} N_i \in \mathcal{N}$. Now,

$$\bigcup_{i=1}^{\infty} (A_i \cup B_i) = \left(\bigcup_{i=1}^{\infty} A_i\right) \cup \left(\bigcup_{i=1}^{\infty} B_i\right), \quad (1.47)$$

where $\bigcup_{i=1}^{\infty} A_i \in \Sigma$, and $\bigcup_{i=1}^{\infty} B_i \subseteq \bigcup_{i=1}^{\infty} N_i \in \mathcal{N}$.

There is a unique extension $\bar{\mu}$ of μ to a complete measure on $\bar{\Sigma}$, given by

$$\bar{\mu}(A \cup B) = \mu(A), \quad (1.48)$$

where $A \in \Sigma$ and $B \subseteq N$ for some $N \in \mathcal{N}$. The measure space $(\Omega, \bar{\Sigma}, \bar{\mu})$ is called the **completion** of (Ω, Σ, μ) .

Proposition 1.7

$\bar{\mu}$ is a complete measure, and it's the unique measure on $\bar{\Sigma}$ that extends μ .

Proof. Suppose $\mu(A \cup B) = 0$ for some $A \cup B \in \bar{\Sigma}$, where $A \in \Sigma$ and $B \subseteq N$ with $\mu(N) = 0$. Choose any subset $C \subseteq A \cup B$. We can write it as

$$C = \emptyset \cup C, \text{ for } C \subseteq A \cup N, \quad (1.49)$$

where $A \cup N \in \Sigma$ with $\mu(A \cup N) = 0$. Therefore, $C \in \bar{\Sigma}$, and hence $\bar{\mu}$ is a complete measure.

Assume for the sake of contradiction that μ' is a measure on $\bar{\Sigma}$ that extends μ , i.e. $\mu'|_{\Sigma} = \mu$. For $A \cup B \in \bar{\Sigma}$, where $A \in \Sigma$ and $B \subseteq N$ with $\mu(N) = 0$, we have

$$\bar{\mu}(A \cup B) = \mu(A) = \mu'(A) \leq \mu'(A \cup B) \leq \mu'(A \cup N) = \mu(A \cup N) = \mu(A) = \bar{\mu}(A \cup B). \quad (1.50)$$

Therefore, $\mu' = \bar{\mu}$. ■

Lemma 1.8

The following implications hold **if and only if** the measure μ is complete:

- (i) if f is measurable and $f = g$ μ -a.e., then g is measurable;
- (ii) if $(f_n)_{n \geq 1}$ are measurable and $f_n \rightarrow f$ μ -a.e., then f is measurable.

Proof. (i) Suppose μ is complete; and f is measurable and $f = g$ a.e. Then $f \neq g$ on $B \subseteq N$, where $\mu(N) = 0$. Since μ is complete, $B \in \Sigma$. Now,

$$\{x \in \Omega : g(x) > a\} = (\{x \in \Omega : g(x) > a\} \cap B) \cup (\{x \in \Omega : g(x) > a\} \cap B^c) \quad (1.51)$$

$$= (\text{subset of a null-set}) \cup (\{x \in \Omega : f(x) > a\} \cap B^c), \quad (1.52)$$

which is a measurable set.

Conversely, suppose that whenever f is measurable and $f = g$ a.e., we have that g is measurable. Now, let $N \in \mathcal{N}$ and $B \subseteq N$. Take

$$f \equiv 0, \quad g = \chi_B \quad (1.53)$$

They agree on N^c , so $f = g$ a.e. f is a constant function so it's measurable. Therefore, g is also measurable, which forces

$$g^{-1}\{1\} = B \in \Sigma. \quad (1.54)$$

(ii) Suppose $f_n \rightarrow g$ (everywhere). Then g is measurable, and $f = g$ a.e. Therefore, f is measurable if and only if μ is complete. ■

Lemma 1.9 (Borel–Cantelli lemma)

Let (Ω, Σ, μ) be a measure space. Let $(E_k)_{k=1}^\infty$ be a countable family of measurable sets of Ω such that

$$\sum_{k=1}^{\infty} \mu(E_k) < \infty. \quad (1.55)$$

Let $E := \{x \in \Omega : x \in E_k \text{ for infinitely many } k\}$. Then E is measurable and $\mu(E) = 0$.

Proof. Note that

$$\begin{aligned} x \in E_k \text{ for infinitely many } k &\iff \forall n \exists m \geq n \text{ such that } x \in E_m \\ &\iff \forall n, x \in \bigcup_{m \geq n} E_m \\ &\iff x \in \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} E_m. \end{aligned}$$

Therefore, $E = \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} E_m$ is measurable as it's made up of measurable sets by countable unions and intersections. Now,

$$\mu\left(\bigcup_{m \geq 1} E_m\right) \leq \sum_{m=1}^{\infty} \mu(E_m) < \infty. \quad (1.56)$$

By continuity, since $\left\{\bigcup_{m \geq n} E_m\right\}_n$ is a decreasing sequence,

$$\mu(E) = \lim_{n \rightarrow \infty} \mu\left(\bigcup_{m \geq n} E_m\right) \leq \lim_{n \rightarrow \infty} \sum_{m=n}^{\infty} \mu(E_m) = 0. \quad (1.57)$$

■

Exercise 1.2. Let $(f_n)_{n \geq 1}$ be a sequence of (measurable) functions on a measure space (Ω, Σ, μ) . Suppose that for every $x \in \Omega$, the sequence $(f_n(x))_{n \geq 1}$ is bounded. Let

$$E := \{x \in \Omega : (f_n(x))_{n \geq 1} \text{ converges}\}. \quad (1.58)$$

Show that $E \in \Sigma$.

Solution. $(f_n(x))_{n \geq 1}$ converges if and only if $\limsup_n f_n(x) = \liminf_n f_n(x)$. We have already shown that $\limsup_n f_n$ and $\liminf_n f_n$ are measurable. Therefore, so is $h := \limsup_n f_n - \liminf_n f_n$. Now,

$$E = \{x \in \Omega : (f_n(x))_{n \geq 1} \text{ converges}\} = h^{-1}\{0\}, \quad (1.59)$$

which is measurable since h is measurable. ■

Exercise 1.3. For each $n \geq 1$, let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Suppose that:

- (i) $0 \leq f_n(x)$ for all n and all $x \in [0, 1]$;
- (ii) the sequence of partial sums $\sum_{n=1}^N f_n(x)$ converges uniformly on $[0, 1]$ as $N \rightarrow \infty$;
- (iii) the sum $\sum_{n=1}^{\infty} (\sup\{f_n(x) : 0 \leq x \leq 1\})$ diverges.

Either give an example of such a sequence of functions, or prove that no such sequence exists.

Solution. One such sequence of functions is

$$f_n(x) = \begin{cases} \frac{1}{n} & \text{if } x = \frac{1}{n}, \\ 0 & \text{otherwise.} \end{cases} \quad (1.60)$$

Then

$$\sum_{n=1}^{\infty} (\sup\{f_n(x) : 0 \leq x \leq 1\}) = \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges.} \quad (1.61)$$

$$\sum_{n=1}^N f_n(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } 1 \leq n \leq N, \\ 0 & \text{otherwise,} \end{cases} \quad (1.62)$$

which converges uniformly to

$$f(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } n = 1, 2, 3, \dots, \\ 0 & \text{otherwise,} \end{cases} \quad (1.63)$$

■

§2 11 September 2026

Having built the machinery of σ -algebras, measures, and measurable functions, we now turn to the central question of this part of the course:

Given a measure space (Ω, Σ, μ) and a measurable function f , how do we define $\int_{\Omega} f \, d\mu$?

The construction proceeds in three stages: first for characteristic functions, then for simple functions, and finally for general non-negative measurable functions (by approximation). Integration of signed or complex-valued functions is deferred to a later lecture.

Definition 2.1 (Characteristic function). Let $A \in \Sigma$ be a measurable set. The **characteristic function** (or **indicator function**) of A is the function $\chi_A : \Omega \rightarrow \mathbb{R}$ (also written $\mathbf{1}_A$) defined by

$$\chi_A(x) := \begin{cases} 1, & x \in A; \\ 0, & x \notin A. \end{cases} \quad (2.1)$$

Remark 2.1. As noted already, χ_A is always measurable when $A \in \Sigma$: for an interval $I \subseteq \mathbb{R}$,

$$\chi_A^{-1}(I) = \begin{cases} \Omega, & 0, 1 \in I; \\ A, & 1 \in I, 0 \notin I; \\ A^c, & 0 \in I, 1 \notin I; \\ \emptyset, & 0, 1 \notin I, \end{cases} \quad (2.2)$$

and each of $\Omega, A, A^c, \emptyset$ lies in Σ .

The entire theory of integration is built by insisting on one basic axiom, motivated by the idea that the integral should compute the “volume” below the graph of a function: for a measurable set A and a constant height $c \geq 0$, the volume of the block of height c over the base A should be c times the size of the base. This gives the defining relation

$$\int_{\Omega} c \chi_A \, d\mu := c \mu(A), \quad c \geq 0, \, A \in \Sigma, \quad (2.3)$$

and, in particular (taking $c = 1$),

$$\int_{\Omega} \chi_A \, d\mu := \mu(A). \quad (2.4)$$

Definition 2.2 (Simple function). A measurable function $\varphi : \Omega \rightarrow \mathbb{R}$ is called **simple** if it takes only finitely many values.

Remark 2.2 (Standard representation). If φ is simple and $a_1, \dots, a_N$ are its distinct *non-zero* values, set

$$A_i := \varphi^{-1}(\{a_i\}), \quad i = 1, \dots, N. \quad (2.5)$$

Each A_i is measurable (since φ is measurable), the sets $A_1, \dots, A_N$ are pairwise disjoint, and

$$\varphi = \sum_{i=1}^N a_i \chi_{A_i}. \quad (2.6)$$

This expression is called the **standard representation** of φ .

Remark 2.3 (Non-standard representations). A simple function can typically be written as a finite linear combination of characteristic functions in more than one way, using sets that are *not* disjoint and coefficients that are *not* distinct. For example, on disjoint sets A, B, C with

$$\varphi = 5\chi_A + 8\chi_B + 3\chi_C \quad (2.7)$$

(the standard representation), one may check directly that

$$\varphi = 5\chi_{A \cup B} + 3\chi_{B \cup C} \quad (2.8)$$

as well, since on A this gives 5, on B it gives $5 + 3 = 8$, and on C it gives 3 — matching the standard representation, even though $A \cup B$ and $B \cup C$ overlap.

Definition 2.3 (Integral of a non-negative simple function). Let φ be a non-negative simple function with standard representation $\varphi = \sum_{i=1}^N a_i \chi_{A_i}$ (so $a_i > 0$ and the A_i are pairwise disjoint, possibly with $\mu(A_i) = \infty$). We define

$$\int_{\Omega} \varphi \, d\mu := \sum_{i=1}^N a_i \mu(A_i) \in [0, \infty]. \quad (2.9)$$

Lemma 2.1 (Independence of the representation)

Let $\varphi \geq 0$ be a simple function, and suppose

$$\varphi = \sum_{n=1}^N a_n \chi_{A_n} = \sum_{m=1}^M b_m \chi_{B_m} \quad (2.10)$$

are two representations of φ as finite linear combinations of characteristic functions, where the sets A_n (resp. B_m) need *not* be pairwise disjoint, and the coefficients $a_n, b_m \in \mathbb{R}$ need *not* be distinct or non-negative (only φ itself is assumed non-negative; we adopt the convention $0 \times \infty = 0$ if some $\mu(A_n) = \infty$ with $a_n = 0$). Then

$$\sum_{n=1}^N a_n \mu(A_n) = \sum_{m=1}^M b_m \mu(B_m). \quad (2.11)$$

Proof. We'll show that $\sum_{n=1}^N a_n \mu(A_n) = \sum_{i=1}^K r_i \mu(E_i)$, where $\varphi = \sum_{i=1}^K r_i \chi_{E_i}$ is the standard representation.

Case 1: A_i 's are pairwise disjoint, but a_i 's need not be distinct. In this case, for each $a \in \{a_1, \dots, a_N\}$, set $B_a = \bigcup_{a_i=a} A_i$. Then it's clear that $B_a \cap B_{a'} = \emptyset$, since A_i 's are pairwise disjoint. Furthermore, $\varphi^{-1}(a) = B_a$. Therefore, $\varphi = \sum_a a \chi_{B_a}$ is the standard representation.

$$\sum_a a \mu(B_a) = \sum_a a \sum_{a_i=a} \mu(A_i) = \sum_{i=1}^N a_i \mu(A_i). \quad (2.12)$$

Case 2: Now let's assume that A_i 's are not necessarily pairwise disjoint.

Claim 2.1: There exists a refinement of the set $\bigcup_{i=1}^N A_i$ with sets $B_1, \dots, B_M$ such that $\bigcup_{i=1}^N A_i = \bigcup_{j=1}^M B_j$, and B_j 's are pairwise disjoint, and for each i , $A_i = \bigcup_{B_j \subseteq A_i} B_j$.

Proof. For each nonempty subset S of $\{1, \dots, N\}$, we set

$$B_S = \left(\bigcap_{i \in S} A_i \right) \cap \left(\bigcap_{i \notin S} A_i^c \right). \quad (2.13)$$

This gives a collection of $2^N - 1$ pairwise disjoint sets. Now, we'll show that

$$A_i = \bigcup_{B_S \subseteq A_i} B_S. \quad (2.14)$$

Clearly, $\bigcup_{B_S \subseteq A_i} B_S \subseteq A_i$ holds. Now, take $x \in A_i$. Let $S' = \{i : x \in A_i\}$, which is a nonempty set. Then $x \in B_{S'} \subseteq A_i$ since S' contains i . Therefore, $\bigcup_{B_S \subseteq A_i} B_S \supseteq A_i$.

This proves $\bigcup_{S \neq \emptyset} B_S \supseteq \bigcup_{i=1}^N A_i$. The reverse containment holds since $B_S \subseteq \bigcup_{i=1}^N A_i$. $\square$

Now, for each S , we define $b_S = \sum_{A_i \supseteq B_S} a_i$. Then

$$\sum_{i=1}^N a_i \mu(A_i) = \sum_{i=1}^N a_i \sum_{B_S \subseteq A_i} \mu(B_S) = \sum_S \sum_{B_S \subseteq A_i} a_i \mu(B_S) = \sum_S b_S \mu(B_S). \quad (2.15)$$

This boils down to case 1, and we know that this is equal to the sum in the standard representation. $\blacksquare$

Remark 2.4. The integral of a non-negative simple function may be computed using *any* representation of φ as a finite linear combination of characteristic functions of measurable sets, not only the standard one. Equivalently, a non-negative simple function may be described as any finite non-negative linear combination of characteristic functions of measurable sets, and its integral is unambiguously

$$\int_{\Omega} \varphi d\mu = \sum_n a_n \mu(A_n) \quad (2.16)$$

for any such representation $\varphi = \sum_n a_n \chi_{A_n}$.

Lemma 2.2

Let φ and ψ be simple functions in $\mathcal{M}^+$.

- (a) If $c \in \mathbb{R}$, then $\int c\varphi \, d\mu = c \int \varphi \, d\mu$.
- (b) $\int (\varphi + \psi) \, d\mu = \int \varphi \, d\mu + \int \psi \, d\mu$.
- (c) If $\varphi \leq \psi$, then $\int \varphi \, d\mu \leq \int \psi \, d\mu$.
- (d) $A \mapsto \int_A \varphi \, d\mu := \int_{\Omega} \varphi \cdot \chi_A \, d\mu$ is a measure on Σ .

Proof. (a) Let $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$. Then $c\varphi = \sum_{i=1}^n ca_i \chi_{A_i}$.

$$\int_{\Omega} c\varphi \, d\mu = \sum_{i=1}^n ca_i \mu(A_i) = c \sum_{i=1}^n a_i \mu(A_i) = c \int_{\Omega} \varphi \, d\mu. \quad (2.17)$$

(b) Let $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$ and $\psi = \sum_{j=1}^m b_j \chi_{B_j}$. Then $\varphi + \psi = \sum_{i=1}^n a_i \chi_{A_i} + \sum_{j=1}^m b_j \chi_{B_j}$.

$$\int_{\Omega} (\varphi + \psi) \, d\mu = \sum_{i=1}^n a_i \mu(A_i) + \sum_{j=1}^m b_j \mu(B_j) = \int_{\Omega} \varphi \, d\mu + \int_{\Omega} \psi \, d\mu. \quad (2.18)$$

(c) $\psi - \varphi$ is a non-negative simple function, whose integral is non-negative, by definition.

(d) Let $\varphi = \sum_{i=1}^n a_i \chi_{E_i}$ be the standard representation.

$$\nu(A) := \int_{\Omega} \varphi \cdot \chi_A \, d\mu = \int_{\Omega} \sum_{i=1}^n a_i \chi_{E_i} \cdot \chi_A \, d\mu = \int_{\Omega} \sum_{i=1}^n a_i \chi_{E_i \cap A} \, d\mu = \sum_{i=1}^n a_i \mu(E_i \cap A). \quad (2.19)$$

Clearly, $\nu(A) \geq 0$ and $\nu(\emptyset) = 0$. Now, if $\{A_n\}_n$ is a countable collection of pairwise disjoint sets,

$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^n a_i \mu\left(E_i \cap \bigcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^n a_i \mu\left(\bigcup_{n=1}^{\infty} (E_i \cap A_n)\right) \quad (2.20)$$

$E_i \cap A_n$'s are all pairwise disjoint, so

$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^n a_i \sum_{n=1}^{\infty} \mu(E_i \cap A_n) = \sum_{n=1}^{\infty} \sum_{i=1}^n a_i \mu(E_i \cap A_n) = \sum_{n=1}^{\infty} \nu(E_i \cap A_n). \quad (2.21)$$

■

We now turn to general non-negative measurable functions. The key technical tool, used repeatedly throughout analysis, is to approximate an object in a large class (here, non-negative measurable functions) by objects in a smaller class we already understand (here, simple functions).

Theorem 2.3 (Approximation by simple functions)

Let $f : \Omega \rightarrow [0, \infty]$ be measurable. Then there exists a sequence of simple functions $(\varphi_n)_{n \geq 1}$, each satisfying $0 \leq \varphi_n \leq \varphi_{n+1}$, such that

$$\varphi_n(x) \longrightarrow f(x) \quad \text{for every } x \in \Omega. \quad (2.22)$$

Proof. For each $n \geq 1$ and each integer k with $0 \leq k \leq n2^n - 1$, define the measurable set

$$A_{k,n} := \left\{ x \in \Omega : \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \right\}, \quad (2.23)$$

and, to handle the top of the range, set

$$A_{n2^n,n} := \{x \in \Omega : f(x) \geq n\}. \quad (2.24)$$

These sets are measurable (being preimages of intervals under f) and, for each fixed n , pairwise disjoint, with

$$\Omega = \bigcup_{k=0}^{n2^n} A_{k,n}. \quad (2.25)$$

Define

$$\varphi_n := \sum_{k=0}^{n2^n} \frac{k}{2^n} \chi_{A_{k,n}}, \quad (2.26)$$

a non-negative simple function. In words: φ_n rounds f down to the nearest multiple of 2^{-n} , on the region where $f < n$, and sets $\varphi_n = n$ where $f \geq n$.

Monotonicity. Passing from level n to level $n+1$ refines each interval $[k/2^n, (k+1)/2^n)$ into

$$\left[\frac{k}{2^n}, \frac{k+1}{2^n} \right) = \left[\frac{2k}{2^{n+1}}, \frac{2k+1}{2^{n+1}} \right) \cup \left[\frac{2k+1}{2^{n+1}}, \frac{2k+2}{2^{n+1}} \right). \quad (2.27)$$

So $A_{k,n} = A_{2k,n+1} \cup A_{2k+1,n+1}$; on either piece, the assigned value at level $n+1$ is at least the value $k/2^n$ assigned at level n . Likewise, the top set $A_{n2^n,n} = \{f \geq n\}$ splits, at level $n+1$, into the part where $n \leq f < n+1$ (now assigned a value $\geq n$, refined further) and the part where $f \geq n+1$ (assigned the value $n+1 > n$). Hence $\varphi_n \leq \varphi_{n+1}$ pointwise.

Pointwise convergence. Fix $x \in \Omega$. If $f(x) = \infty$, then $\varphi_n(x) = n \rightarrow \infty = f(x)$. If $f(x) < \infty$, then once $n > f(x)$, we have $x \in A_{k,n}$ for the unique k with

$$\frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}, \quad (2.28)$$

so

$$0 \leq f(x) - \varphi_n(x) < \frac{1}{2^n} \longrightarrow 0. \quad (2.29)$$

Hence $\varphi_n(x) \rightarrow f(x)$ for every $x \in \Omega$. ■

Remark 2.5. The convergence in this proposition holds **everywhere**, not merely almost everywhere, and the approximating sequence is **monotonically increasing**. Both features will be essential in the proof of the Monotone Convergence Theorem below.

Definition 2.4 (Integral of a non-negative measurable function). Let $f : \Omega \rightarrow [0, \infty]$ be measurable. We define

$$\int_{\Omega} f \, d\mu := \sup \left\{ \int_{\Omega} \varphi \, d\mu : \varphi \text{ simple, } 0 \leq \varphi \leq f \right\} \in [0, \infty]. \quad (2.30)$$

Remark 2.6. This definition could have been adopted directly, without first proving the approximation proposition above; the proposition simply confirms that the class of simple functions lying below f is rich enough to be useful (e.g. in proving the Monotone Convergence Theorem).

Notation for the integral varies across the literature and is otherwise a matter of convention: one may write $\int_{\Omega} f \, d\mu$, $\int_{\Omega} f(x) \, d\mu(x)$, $\int f \, d\mu$ (when Ω is clear), or $\int f$ (when μ is also clear); some authors (especially in physics) write the measure before the function. Any of these is acceptable as long as the underlying set, σ -algebra, and measure are unambiguous.

The following two properties follow immediately from the definition as a supremum.

Proposition 2.4 (Monotonicity)

If $f, g : \Omega \rightarrow [0, \infty]$ are measurable and $f \leq g$, then

$$\int_{\Omega} f \, d\mu \leq \int_{\Omega} g \, d\mu. \quad (2.31)$$

Proof. Every simple function φ with $0 \leq \varphi \leq f$ also satisfies $0 \leq \varphi \leq g$, so the set over which the supremum defining $\int_{\Omega} f \, d\mu$ is taken is contained in the set over which the supremum defining $\int_{\Omega} g \, d\mu$ is taken. A supremum over a smaller set cannot exceed a supremum over a larger set. ■

Proposition 2.5 (Positive homogeneity)

If $f : \Omega \rightarrow [0, \infty]$ is measurable and $c \geq 0$, then

$$\int_{\Omega} cf \, d\mu = c \int_{\Omega} f \, d\mu. \quad (2.32)$$

Proof. This conclusion clearly holds if $c = 0$. For $c > 0$, note that $\varphi \leq cf \iff \frac{\varphi}{c} \leq f$. ■

Remark 2.7. Additivity of the integral, $\int_{\Omega} (f + g) \, d\mu = \int_{\Omega} f \, d\mu + \int_{\Omega} g \, d\mu$ for non-negative measurable f, g , is *not* immediate from the definition and requires a genuine proof; this will follow once the Monotone Convergence Theorem is available (approximate f and g by increasing sequences of simple functions, for which additivity is straightforward, and pass to the limit).

Definition 2.5 (Integral over a subset). Let $A \in \Sigma$ and let $g : \Omega \rightarrow [0, \infty]$ be measurable. The integral of g over A is defined by

$$\int_A g \, d\mu := \int_{\Omega} g \chi_A \, d\mu. \quad (2.33)$$

We now come to the first of the two fundamental convergence theorems of integration theory (the second being the Dominated Convergence Theorem, to be treated in a later lecture). Both address the same basic question: when can a limit be passed inside an integral?

Theorem 2.6 (Monotone Convergence Theorem)

Let $(f_n)_{n \geq 1}$ be a sequence of measurable functions $\Omega \rightarrow [0, \infty]$ such that

$$0 \leq f_n \leq f_{n+1} \quad \text{for every } n, \quad (2.34)$$

and let $f(x) := \lim_{n \rightarrow \infty} f_n(x)$ (which exists in $[0, \infty]$ for every x , by monotonicity). Then f is measurable, and

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (2.35)$$

Proof. Since $(f_n(x))_n$ is non-decreasing for each x , its limit equals $\limsup_n f_n(x) = \sup_n f_n(x)$, which is measurable by .

For each n , since $f_n \leq f$, monotonicity of the integral gives

$$\int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu \quad \text{for all } n. \quad (\star)$$

The left-hand side is non-decreasing in n (again by monotonicity, since $f_n \leq f_{n+1}$) and bounded above by the fixed quantity $\int_{\Omega} f \, d\mu$, so the limit exists and

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu. \quad (2.36)$$

Fix a simple function φ with $0 \leq \varphi \leq f$, and fix a constant $\alpha \in (0, 1)$. Define

$$A_n := \{x \in \Omega : f_n(x) \geq \alpha \varphi(x)\}. \quad (2.37)$$

Each A_n is measurable (as $f_n - \alpha \varphi$ is measurable).

Claim 2.2: $A_n \subseteq A_{n+1}$ for every n .

Proof. If $f_n(x) \geq \alpha \varphi(x)$, then since $f_{n+1}(x) \geq f_n(x)$, also $f_{n+1}(x) \geq \alpha \varphi(x)$. $\square$

Claim 2.3: $\bigcup_{n=1}^{\infty} A_n = \Omega$.

Proof. Fix $x \in \Omega$. If $\varphi(x) = 0$, then $x \in A_1$ trivially (as $f_1(x) \geq 0 = \alpha\varphi(x)$). If $\varphi(x) > 0$, then

$$\alpha\varphi(x) < \varphi(x) \leq f(x). \quad (2.38)$$

Since $f_n(x) \rightarrow f(x)$, eventually $f_n(x) > \alpha\varphi(x)$, i.e. $x \in A_n$ for n large. $\square$

By monotonicity and the definition of the integral over a subset,

$$\int_{\Omega} f_n \, d\mu \geq \int_{A_n} f_n \, d\mu \geq \int_{A_n} \alpha\varphi \, d\mu = \alpha \int_{A_n} \varphi \, d\mu. \quad (2.39)$$

Since φ is simple, the set function $B \mapsto \int_B \varphi \, d\mu$ is a measure. Since (A_n) is an increasing sequence of measurable sets with $\bigcup_n A_n = \Omega$, by [Proposition 1.4](#),

$$\lim_{n \rightarrow \infty} \int_{A_n} \varphi \, d\mu = \int_{\Omega} \varphi \, d\mu. \quad (2.40)$$

Taking $n \rightarrow \infty$ in (2.39),

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \alpha \int_{\Omega} \varphi \, d\mu. \quad (2.41)$$

This holds for every $\alpha \in (0, 1)$; letting $\alpha \rightarrow 1^-$ gives

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \int_{\Omega} \varphi \, d\mu. \quad (2.42)$$

Finally, taking the supremum over all simple φ with $0 \leq \varphi \leq f$,

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \int_{\Omega} f \, d\mu. \quad (2.43)$$

Combining both inequalities gives the result. $\blacksquare$

Remark 2.8. The restriction $\alpha < 1$ (rather than simply taking $\alpha = 1$, i.e. $A_n = \{f_n \geq \varphi\}$) is essential: without it, claim (b) above can fail. It is possible for a simple function $\varphi \leq f$ to touch f exactly at some point x , while the approximating sequence $f_n(x)$ increases to $f(x)$ without ever reaching it, so that $f_n(x) < \varphi(x)$ for every finite n .

Remark 2.9. The [Monotone Convergence Theorem](#) is, in essence, already implicit in the construction of $\int_{\Omega} f \, d\mu$. The integral of f was defined as a supremum of integrals of simple functions below f , and we exhibited an increasing sequence of simple functions converging to f everywhere. The [Monotone Convergence Theorem](#) shows that this phenomenon persists for *any* increasing sequence of non-negative measurable functions converging to f , not only sequences of simple functions.

Lemma 2.7

If $f_1, f_2 \in \mathcal{M}^+$, then $\int_{\Omega} (f_1 + f_2) d\mu = \int_{\Omega} f_1 d\mu + \int_{\Omega} f_2 d\mu$. By induction,

$$\int_{\Omega} \sum_{i=1}^n f_i d\mu = \sum_{i=1}^n \int_{\Omega} f_i d\mu. \quad (2.44)$$

Proof. By [Theorem 2.3](#), we can find increasing sequence of simple functions $\varphi_n \rightarrow f_1, \psi_n \rightarrow f_2$. Then $\varphi_n + \psi_n$ is an increasing sequence of simple function that monotonically converges to $f_1 + f_2$. By [Monotone Convergence Theorem](#),

$$\int_{\Omega} (f_1 + f_2) d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} (\varphi_n + \psi_n) d\mu \quad (2.45)$$

$$= \lim_{n \rightarrow \infty} \int_{\Omega} \varphi_n d\mu + \lim_{n \rightarrow \infty} \int_{\Omega} \psi_n d\mu \quad (2.46)$$

$$= \int_{\Omega} f_1 d\mu + \int_{\Omega} f_2 d\mu. \quad (2.47)$$

■

Proposition 2.8

If $\{f_n\}_n$ is a sequence in $\mathcal{M}^+$, then

$$\int_{\Omega} \sum_{i=1}^{\infty} f_i d\mu = \sum_{i=1}^{\infty} \int_{\Omega} f_i d\mu. \quad (2.48)$$

Proof. $g_n = \sum_{i=1}^n f_i$ is a monotone increasing sequence converging to $f = \sum_{i=1}^{\infty} f_i$. Therefore, by [Monotone Convergence Theorem](#),

$$\int_{\Omega} \sum_{i=1}^{\infty} f_i d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} g_n d\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^n \int_{\Omega} f_i d\mu = \sum_{i=1}^{\infty} \int_{\Omega} f_i d\mu. \quad (2.49)$$

■

Proposition 2.9

If $f \in \mathcal{M}^+$ and $\int_{\Omega} f d\mu = 0$, then $f = 0$ μ -a.e.

Proof. Suppose $f = 0$ a.e. Let E be the set where f is nonzero, then $\mu(E) = 0$ by hypothesis. Then consider a simple function in its standard form $\varphi = \sum_{i=1}^n a_i \chi_{A_i} \leq f$, with $a_i > 0$ and A_i 's pairwise disjoint. Since $\varphi \leq f$, hence each $A_i \subseteq E$, and consequently, $\mu(A_i) = 0$. Then

$$\int_{\Omega} \varphi d\mu = \sum_{i=1}^n a_i \mu(A_i) = 0 \implies \int_{\Omega} f d\mu = \sup_{\substack{0 \leq \varphi \leq f \\ \varphi \text{ simple}}} \int_{\Omega} \varphi d\mu = 0. \quad (2.50)$$

Conversely, suppose $\int_{\Omega} f \, d\mu = 0$. Consider the set

$$E_n = \left\{ x \in \Omega : f(x) > \frac{1}{n} \right\}. \quad (2.51)$$

Then we have

$$E := \{x \in \Omega : f(x) > 0\} = \bigcup_{n=1}^{\infty} E_n. \quad (2.52)$$

We need to show that $\mu(E) = 0$. Assume for the sake of contradiction that it's strictly greater than 0. By continuity ([Proposition 1.4](#)),

$$0 < \mu(E) = \lim_{n \rightarrow \infty} \mu(E_n), \quad (2.53)$$

implies that there exists some m for which $\mu(E_m) > 0$. Then $0 \leq \frac{1}{m} \chi_{E_m} \leq f$, but then

$$\int_{\Omega} f \, d\mu \geq \int_{\Omega} \frac{1}{m} \chi_{E_m} \, d\mu = \frac{1}{m} \mu(E_m) > 0. \quad (2.54)$$

■

Theorem 2.10

If $\{f_n\}_{n \geq 1}$ and f are functions in $\mathcal{M}^+$ such that $\{f_n\}_{n \geq 1}$ converges monotonically increasing to f for μ -a.e. $x \in \Omega$. Then conclusion of [Monotone Convergence Theorem](#) still holds, i.e.

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (2.55)$$

Proof. Suppose $f_n(x) \rightarrow f(x)$ monotonically increasing on $E \subseteq \Omega$, then $\mu(E^c) = 0$. Then $f - f\chi_E = 0$ and $f_n - f_n\chi_E = 0$ a.e. Also, $f_n\chi_E$ converges monotonically increasing to $f\chi_E$ everywhere. As a result, by [Monotone Convergence Theorem](#),

$$\int_{\Omega} f \, d\mu = \int_{\Omega} f\chi_E \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n\chi_E \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (2.56)$$

■

Proposition 2.11

If $f \in \mathcal{M}^+$, then the map

$$A \mapsto \int_A f \, d\mu = \int_{\Omega} f \cdot \chi_A \, d\mu \quad (2.57)$$

is a measure on Σ .

Proof. Suppose $\nu(A) = \int_{\Omega} f \cdot \chi_A \, d\mu$. Clearly, $\nu(A) \geq 0$. Also

$$\nu(\emptyset) = \int_{\Omega} f \cdot \chi_{\emptyset} \, d\mu = \int_{\Omega} 0 \, d\mu = 0. \quad (2.58)$$

Now, suppose $\{A_n\}_{n \geq 1}$ is a countable collection of pairwise disjoint sets. We consider

$$f_n = \sum_{i=1}^n (f \chi_{A_i}) = f \chi_{\bigcup_{i=1}^n A_i}, \quad (2.59)$$

which is a monotonically increasing sequence of measurable functions, that converge pointwise to $f \chi_{\bigcup_{i=1}^{\infty} A_i}$. Then by [Monotone Convergence Theorem](#),

$$\nu \left(\bigcup_{i=1}^{\infty} A_i \right) = \int_{\Omega} f \chi_{\bigcup_{i=1}^{\infty} A_i} d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n d\mu \quad (2.60)$$

$$= \lim_{n \rightarrow \infty} \int_{\Omega} \sum_{i=1}^n (f \chi_{A_i}) d\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^n \int_{\Omega} (f \chi_{A_i}) d\mu \quad (2.61)$$

$$= \sum_{i=1}^{\infty} \nu(A_i). \quad (2.62)$$

■

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In our last lecture, we proved that if f_n is a monotonically increasing sequence of measurable functions, then we can essentially swap limit and integral:

$$\int_{\Omega} \lim_{n \rightarrow \infty} f_n \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.1)$$

The hypothesis that (f_n) is an *increasing* sequence is essential, as the following examples show.

Example 3.1

Consider $f_n = \chi_{(n, n+1)}$. Then $f_n(x) \rightarrow 0$ pointwise for every $x \in \mathbb{R}$, yet

$$\int_{\mathbb{R}} f_n \, d\mu = 1 \quad \text{for every } n, \quad \text{while} \quad \int_{\mathbb{R}} \left(\lim_{n \rightarrow \infty} f_n \right) \, d\mu = 0. \quad (3.2)$$

So the conclusion of [Monotone Convergence Theorem](#) fails without monotonicity.

Example 3.2

Define, for $n \geq 1$,

$$f_n = n \chi_{(0, \frac{1}{n})}, \quad (3.3)$$

or, to make it continuous, a triangular spike on $[0, 1/n]$ of height $2n$ so that its graph is a triangle of area 1. Then

$$\int_0^1 f_n(x) \, dx = 1 \quad \text{for every } n, \quad (3.4)$$

but for every fixed $x > 0$ there is N such that $1/n < x$ for all $n \geq N$, so $f_n(x) \rightarrow 0$ pointwise (and $f_n(0) = 0$ for all n , by construction). Hence $f_n \rightarrow 0$ pointwise while $\int_0^1 f_n \, d\mu = 1$, again showing that monotonicity cannot be dropped.

This type of example is the classical illustration that pointwise convergence alone does not let one pass the limit inside the integral; one really needs either monotonicity or a dominating integrable function ([Dominated Convergence Theorem](#), see below).

Theorem 3.1 (Fatou's Lemma)

Let (f_n) be a sequence of nonnegative measurable functions on Ω . Then

$$\int_{\Omega} \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.5)$$

Proof. For each m , define

$$g_m(x) = \inf_{n \geq m} f_n(x). \quad (3.6)$$

For every $n \geq m$ we have $g_m \leq f_n$, and since the integral is monotone,

$$\int g_m \, d\mu \leq \int f_n \, d\mu \quad \text{for all } n \geq m. \quad (3.7)$$

Taking the infimum over $n \geq m$ on the right-hand side gives

$$\int g_m \, d\mu \leq \inf_{n \geq m} \int f_n \, d\mu. \quad (3.8)$$

The sequence (g_m) is increasing in m (an infimum over a smaller index set is larger), and

$$\lim_{m \rightarrow \infty} g_m(x) = \sup_m \inf_{n \geq m} f_n(x) = \liminf_{n \rightarrow \infty} f_n(x). \quad (3.9)$$

By the [Monotone Convergence Theorem](#) applied to (g_m) ,

$$\int \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu = \lim_{m \rightarrow \infty} \int g_m \, d\mu \leq \lim_{m \rightarrow \infty} \inf_{n \geq m} \int f_n \, d\mu = \liminf_{n \rightarrow \infty} \int f_n \, d\mu. \quad (3.10)$$

■

Proposition 3.2

If $\{f_n\}_{n \geq 1}$ and f are functions in $\mathcal{M}^+$ such that $f_n(x)$ converges to $f(x)$ for μ -a.e. $x \in \Omega$, then the conclusion of [Fatou's Lemma](#) still holds, i.e.

$$\int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.11)$$

Proof. Suppose $f_n(x) \rightarrow f(x)$ on $E \subseteq \Omega$, then $\mu(E^c) = 0$. Then $f - f\chi_E = 0$ and $f_n - f_n\chi_E = 0$ a.e. Also, $f_n\chi_E$ converges to $f\chi_E$ everywhere. As a result,

$$\int_{\Omega} f \, d\mu = \int_{\Omega} f\chi_E \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n\chi_E \, d\mu = \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.12)$$

■

Proposition 3.3

If $f \in \mathcal{M}^+$ and $\int_{\Omega} f \, d\mu < \infty$, then $\{x \in \Omega : f(x) = \infty\}$ is a null set and $\{x \in \Omega : f(x) > 0\}$ is σ -finite.

Proof. Suppose $E := \{x \in \Omega : f(x) = \infty\}$. Assume for the sake of contradiction that $\mu(E) > 0$. Then $f \geq f \cdot \chi_E$. So

$$\int_{\Omega} f \, d\mu \geq \int_{\Omega} f \cdot \chi_E \, d\mu = \int_{\Omega} \infty \cdot \chi_E \, d\mu = \infty \cdot \mu(E) = \infty, \quad (3.13)$$

which is a contradiction! Now, suppose $F = \{x \in \Omega : f(x) > 0\}$. Observe that

$$F = \bigcup_{n=1}^{\infty} \left\{ x \in \Omega : f(x) \geq \frac{1}{n} \right\} =: \bigcup_{n=1}^{\infty} F_n. \quad (3.14)$$

So it suffices to prove that each F_n has finite measure. Note that $f(x) \geq \frac{1}{n}$ on F_n , so that $n \cdot f(x) \geq 1$. In other words, $\chi_{F_n} \leq n \cdot f$. By monotonicity of integrals,

$$\mu(F_n) = \int_{\Omega} \chi_{F_n} d\mu \leq \int_{\Omega} n \cdot f d\mu = n \int_{\Omega} f d\mu < \infty. \quad (3.15)$$

■

Definition 3.1 (Positive and negative part). For a measurable function $f : \Omega \rightarrow \mathbb{R}$ define

$$f^+(x) = \max\{f(x), 0\}, \quad f^-(x) = \max\{-f(x), 0\}, \quad (3.16)$$

to be the **positive part** and **negative part** of f , respectively.

Remark 3.1. It's clear that $f^+, f^- \geq 0$ are measurable, and

$$f = f^+ - f^-, \quad |f| = f^+ + f^-. \quad (3.17)$$

This decomposition is not unique as a way of writing f as a difference of two nonnegative functions (one may add the same nonnegative function to both f^+ and f^-), but it is the canonical choice, and it is the one used to define the integral of a general function.

Definition 3.2 (The space L^1). Let (Ω, μ) be a measure space. Define

$$L^1(\Omega, \mu) = \left\{ f : \Omega \rightarrow \mathbb{R} \text{ measurable} : \int_{\Omega} |f| d\mu < \infty \right\}. \quad (3.18)$$

For $f \in L^1(\Omega, \mu)$, both $\int f^+ d\mu$ and $\int f^- d\mu$ are finite (since their sum is $\int |f| d\mu < \infty$), and we define

$$\int_{\Omega} f d\mu := \int_{\Omega} f^+ d\mu - \int_{\Omega} f^- d\mu. \quad (3.19)$$

Remark 3.2. The integral is first defined for simple functions, then extended to nonnegative measurable functions via the supremum definition, and finally extended to L^1 via this difference, precisely so as to avoid ever having to make sense of $\infty - \infty$.

Proposition 3.4

If $f \in L^1$, then

$$\left| \int_{\Omega} f d\mu \right| \leq \int_{\Omega} |f| d\mu. \quad (3.20)$$

Proof.

$$\left| \int_{\Omega} f \, d\mu \right| = \left| \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu \right| \leq \int_{\Omega} f^+ \, d\mu + \int_{\Omega} f^- \, d\mu = \int_{\Omega} |f| \, d\mu. \quad (3.21)$$

■

Proposition 3.5

If $f, g \in L^1(\Omega, \mu)$ and $c \in \mathbb{R}$, then $f + cg \in L^1(\Omega, \mu)$ and

$$\int_{\Omega} (f + cg) \, d\mu = \int_{\Omega} f \, d\mu + c \int_{\Omega} g \, d\mu. \quad (3.22)$$

Proof. First notice that

$$|f + cg| \leq |f| + |c| |g|, \quad (3.23)$$

which implies using monotonicity of integral of non-negative functions that

$$\int_{\Omega} |f + cg| \, d\mu \leq \int_{\Omega} |f| + |c| |g| \, d\mu = \int_{\Omega} |f| \, d\mu + |c| \int_{\Omega} |g| \, d\mu < \infty. \quad (3.24)$$

We'll now show that $\int_{\Omega} cf \, d\mu = c \int_{\Omega} f \, d\mu$. For $c \geq 0$,

$$(cf)^+ = cf^+, (cf)^- = cf^-. \quad (3.25)$$

Then

$$\int_{\Omega} cf \, d\mu = \int_{\Omega} (cf)^+ \, d\mu - \int_{\Omega} (cf)^- \, d\mu = \int_{\Omega} cf^+ \, d\mu - \int_{\Omega} cf^- \, d\mu = c \int_{\Omega} f \, d\mu. \quad (3.26)$$

For $c < 0$,

$$(cf)^+ = -cf^-, (cf)^- = -cf^+. \quad (3.27)$$

Then

$$\int_{\Omega} cf \, d\mu = \int_{\Omega} (cf)^+ \, d\mu - \int_{\Omega} (cf)^- \, d\mu = \int_{\Omega} -cf^- \, d\mu - \int_{\Omega} -cf^+ \, d\mu = c \int_{\Omega} f \, d\mu. \quad (3.28)$$

Now let's show that $\int_{\Omega} f + g \, d\mu = \int_{\Omega} f \, d\mu + \int_{\Omega} g \, d\mu$. Suppose $h = f + g$. Then

$$h^+ - h^- = f^+ - f^- + g^+ - g^- \implies h^+ + f^- + g^- = h^- + f^+ + g^+. \quad (3.29)$$

Using the additivity of integral of non-negative functions, we get

$$\int_{\Omega} h^+ \, d\mu + \int_{\Omega} f^- \, d\mu + \int_{\Omega} g^- \, d\mu = \int_{\Omega} h^- \, d\mu + \int_{\Omega} f^+ \, d\mu + \int_{\Omega} g^+ \, d\mu. \quad (3.30)$$

Regrouping gives us the desired result:

$$\int_{\Omega} h^+ \, d\mu - \int_{\Omega} h^- \, d\mu = + \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu + \int_{\Omega} g^+ \, d\mu - \int_{\Omega} g^- \, d\mu, \quad (3.31)$$

i.e. $\int_{\Omega} h \, d\mu = \int_{\Omega} f \, d\mu + \int_{\Omega} g \, d\mu$. ■

Lemma 3.6

If $f \leq g$ a.e. then $\int_{\Omega} f \, d\mu \leq \int_{\Omega} g \, d\mu$.

Proof. Suppose $f \leq g$ on E with $\mu(E^c) = 0$. Then $(g - f)\chi_E \geq 0$. The integral of a non-negative function is non-negative, so

$$\int_{\Omega} g \, d\mu - \int_{\Omega} f \, d\mu = \int_{\Omega} (g - f) \, d\mu = \int_{\Omega} (g - f)\chi_E \, d\mu \geq 0. \quad (3.32)$$
■

Proposition 3.7

If $f \in M$ and $g \in L^1(\Omega, \mu)$ with $|f| \leq g$ a.e., then $f \in L^1(\Omega, \mu)$. (A function dominated by an integrable function is itself integrable.)

Proof. By monotonicity of integrals,

$$\int_{\Omega} |f| \, d\mu \leq \int_{\Omega} g \, d\mu < \infty. \quad (3.33)$$
■

Proposition 3.8

If $f, g \in L^1$, then the following are equivalent:

- (a) $\int_A f \, d\mu = \int_A g \, d\mu$ for every $A \in \Sigma$;
- (b) $\int_{\Omega} |f - g| \, d\mu = 0$;
- (c) $f = g$ a.e.

Proof. The equivalence of (b) and (c) follows from [Proposition 2.9](#):

$$\int_{\Omega} |f - g| \, d\mu = 0 \iff |f - g| = 0 \text{ a.e.} \iff f = g \text{ a.e.} \quad (3.34)$$

(b) $\Rightarrow$ (a) Take any $A \in \Sigma$. Then

$$\left| \int_A f \, d\mu - \int_A g \, d\mu \right| = \left| \int_{\Omega} (f - g)\chi_A \, d\mu \right| \leq \int_{\Omega} |f - g|\chi_A \, d\mu \leq \int_{\Omega} |f - g| \, d\mu = 0. \quad (3.35)$$

(a) $\Rightarrow$ (c) Suppose $A = \{x \in \Omega : f(x) > g(x)\}$, which is a measurable set. On A , $f - g > 0$. If $\mu(A) > 0$, then $\int_A (f - g) > 0$ by the following claim, which contradicts with the hypothesis.

Claim 3.1: If $\mu(E) > 0$ and $h > 0$ on E , then $\int_E h \, d\mu > 0$.

Proof. Let $E_n = \{x \in E : h(x) \geq \frac{1}{n}\}$. Then $\{E_n\}_n$ is an increasing sequence with

$$E = \bigcup_{i=1}^{\infty} E_n. \quad (3.36)$$

By continuity (Proposition 1.4),

$$0 < \mu(E) = \lim_{n \rightarrow \infty} \mu(E_n), \quad (3.37)$$

implies that there exists some m for which $\mu(E_m) > 0$. Then $0 \leq \frac{1}{m} \chi_{E_m} \leq h$, but then

$$\int_E h \, d\mu \geq \int_{\Omega} \frac{1}{m} \chi_{E_m} \, d\mu = \frac{1}{m} \mu(E_m) > 0. \quad (3.38)$$

□

Therefore, $\mu(A) = 0$. Similarly, if $B = \{x \in \Omega : f(x) < g(x)\}$, then $\mu(B) = 0$. Now,

$$\{x \in \Omega : f(x) \neq g(x)\} = A \cup B, \quad (3.39)$$

is also a null set, since it's the union of two null sets. ■

Remark 3.3. Note that $\int_{\Omega} f \, d\mu = \int_{\Omega} g \, d\mu$ does not necessarily imply that $f = g$ a.e. For example, consider $\Omega = [0, 2\pi]$ and $f(x) = \sin x$, $g(x) = \cos x$. Then

$$\int_0^{2\pi} \sin x \, dx = \int_0^{2\pi} \cos x \, dx = 0, \quad (3.40)$$

but it's **NOT** true that $\sin x = \cos x$ a.e.

Theorem 3.9 (Dominated Convergence Theorem)

Let (f_n) be a sequence of measurable functions on Ω such that $f_n \rightarrow f$ pointwise a.e., and suppose there is a function $g \in L^1(\Omega, \mu)$ such that

$$|f_n(x)| \leq g(x) \text{ a.e. for all } n. \quad (3.41)$$

Then $f \in L^1(\Omega, \mu)$ and

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.42)$$

Proof. Since $|f_n| \leq g$ for every n and $f_n \rightarrow f$ pointwise, also $|f| \leq g$, so $f \in L^1(\Omega, \mu)$ Proposition 3.7.

Since $g + f_n \geq 0$ (because $|f_n| \leq g$) and $g + f_n \rightarrow g + f$ pointwise a.e., [Fatou's Lemma](#) gives

$$\int_{\Omega} (g + f) \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g + f_n) \, d\mu = \int_{\Omega} g \, d\mu + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.43)$$

Since $\int_{\Omega} g \, d\mu$ is finite, it may be cancelled from both sides:

$$\int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.44)$$

Likewise $g - f_n \geq 0$ and $g - f_n \rightarrow g - f$ pointwise a.e., so [Fatou's Lemma](#) gives

$$\int_{\Omega} (g - f) \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g - f_n) \, d\mu = \int_{\Omega} g \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu, \quad (3.45)$$

using $\liminf(-a_n) = -\limsup(a_n)$. Cancelling $\int_{\Omega} g \, d\mu$ (finite) from both sides:

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu. \quad (3.46)$$

Combining (3.44) and (3.46),

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu, \quad (3.47)$$

and since always $\liminf \leq \limsup$, all three quantities coincide. Hence $\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu$ exists and equals $\int_{\Omega} f \, d\mu$. ■

Remark 3.4. The [Monotone Convergence Theorem](#), [Fatou's Lemma](#), and the [Dominated Convergence Theorem](#) are the three fundamental convergence theorems needed to justify moving a limit inside an integral: either the sequence is monotone (Monotone Convergence), or it is dominated by a single integrable function (Dominated Convergence); [Fatou's Lemma](#) always applies to nonnegative sequences and is the tool used to prove the other two.

Theorem 3.10 (Generalized Dominated Convergence Theorem)

If $f_n, g_n, g \in L^1$, $f_n \rightarrow f$ and $g_n \rightarrow g$ a.e.,

$$|f_n| \leq g_n, \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu = \int_{\Omega} g \, d\mu, \quad (3.48)$$

then $f \in L^1$, and

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu = \int_{\Omega} f \, d\mu. \quad (3.49)$$

Proof. Since $|f_n| \leq g_n$ for every n and $f_n \rightarrow f$ pointwise a.e. and $g_n \rightarrow g$ a.e., we have that $|f| \leq g$ a.e. So $f \in L^1(\Omega, \mu)$ [Proposition 3.7](#).

Since $g_n + f_n \geq 0$ (because $|f_n| \leq g_n$) and $g_n + f_n \rightarrow g + f$ pointwise a.e., [Fatou's Lemma](#) gives

$$\begin{aligned}
\int_{\Omega} g \, d\mu + \int_{\Omega} f \, d\mu &= \int_{\Omega} (g + f) \, d\mu \\
&\leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g_n + f_n) \, d\mu \\
&= \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \\
&= \int_{\Omega} g \, d\mu + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu.
\end{aligned} \tag{3.50}$$

Since $\int_{\Omega} g \, d\mu$ is finite, it may be cancelled from both sides:

$$\int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \tag{3.51}$$

Likewise $g_n - f_n \geq 0$ and $g_n - f_n \rightarrow g - f$ pointwise a.e., so [Fatou's Lemma](#) gives

$$\begin{aligned}
\int_{\Omega} g \, d\mu - \int_{\Omega} f \, d\mu &= \int_{\Omega} (g - f) \, d\mu \\
&\leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g_n - f_n) \, d\mu \\
&= \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \\
&= \int_{\Omega} g \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu.
\end{aligned} \tag{3.52}$$

Cancelling $\int_{\Omega} g \, d\mu$ (finite) from both sides:

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu. \tag{3.53}$$

Combining (3.51) and (3.53),

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu, \tag{3.54}$$

and since always $\liminf \leq \limsup$, all three quantities coincide. Hence $\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu$ exists and equals $\int_{\Omega} f \, d\mu$. ■

Remark 3.5. In the proof of [Generalized Dominated Convergence Theorem](#), we used the fact that if a_n, b_n are two sequences such that $a_n \rightarrow a$, then

$$\liminf_{n \rightarrow \infty} (a_n + b_n) = a + \liminf_{n \rightarrow \infty} b_n, \tag{3.55}$$

$$\liminf_{n \rightarrow \infty} (a_n - b_n) = a - \limsup_{n \rightarrow \infty} b_n. \tag{3.56}$$

To prove the first one, notice that

$$\inf_{m \geq n} (a_m + b_m) \geq \inf_{m \geq n} a_m + \inf_{m \geq n} b_m. \tag{3.57}$$

Taking $n \rightarrow \infty$, we get

$$\liminf_{n \rightarrow \infty} (a_n + b_n) \geq \liminf_{n \rightarrow \infty} a_n + \liminf_{n \rightarrow \infty} b_n \quad (3.58)$$

$$= a + \liminf_{n \rightarrow \infty} b_n. \quad (3.59)$$

On the other hand, writing $b_n = (a_n + b_n) + (-a_n)$ and applying (3.58), we get

$$\liminf_{n \rightarrow \infty} b_n = \liminf_{n \rightarrow \infty} [(a_n + b_n) + (-a_n)] \quad (3.60)$$

$$\geq \liminf_{n \rightarrow \infty} (a_n + b_n) + \liminf_{n \rightarrow \infty} (-a_n) \quad (3.61)$$

$$= \liminf_{n \rightarrow \infty} (a_n + b_n) - a. \quad (3.62)$$

Therefore, combining (3.59) and (3.62), we have established (3.55). To see (3.56),

$$\liminf_{n \rightarrow \infty} (a_n - b_n) = a + \liminf_{n \rightarrow \infty} (-b_n) = a - \limsup_{n \rightarrow \infty} b_n.$$

Example 3.3

Let

$$f_n(x) = n \chi_{(0,1/n)}(x), \quad x \in [0, 1]. \quad (3.63)$$

Then $f_n(x) \rightarrow 0$ for every $x \in [0, 1]$ (for $x = 0$ every $f_n(0) = 0$; for $x > 0$, eventually $\frac{1}{n} < x$), so $f_n \rightarrow 0$ pointwise. However,

$$\int_0^1 f_n dx = 1 \quad \text{for every } n, \quad \text{while } \int_0^1 \left(\lim_{n \rightarrow \infty} f_n \right) dx = 0. \quad (3.64)$$

The conclusion of the Dominated Convergence Theorem fails here because no single integrable function g dominates all the f_n . The pointwise supremum of the family is essentially $g(x) = \frac{1}{x}$, and

$$\int_0^1 \frac{1}{x} dx = \infty, \quad (3.65)$$

so $g \notin L^1([0, 1])$. This also shows the Monotone Convergence Theorem does not apply, since the sequence (f_n) is not pointwise increasing.

We now show that one cannot hope to define a measure on *all* subsets of $\mathbb{R}$ that

- (i) is translation invariant,
- (ii) is monotone (if $A \subseteq B$ then $\mu(A) \leq \mu(B)$), and
- (iii) and that extends the usual length of intervals.

Consequently the σ -algebra of measurable sets must be a strict subset of the power set $\mathcal{P}(\mathbb{R})$. For this, we illustrate the following example:

On $[0, 1]$, declare $x \sim y$ if $x - y \in \mathbb{Q}$. This is an equivalence relation, partitioning $[0, 1]$ into equivalence classes. By the Axiom of Choice, choose exactly one representative x_I from each

equivalence class I , and let

$$A = \{x_I : I \in [0, 1] / \sim\} \subseteq [0, 1], \quad (3.66)$$

where I indexes the set of equivalence classes. The set A is called a *Vitali set*.

Theorem 3.11

There is no measure μ on $\mathcal{P}(\mathbb{R})$ that is translation invariant, monotone, and assigns to every interval its usual length.

Proof. Suppose such a μ exists. For $r \in \mathbb{Q} \cap [-1, 1]$ set

$$Y_r = A + r = \{a + r : a \in A\}. \quad (3.67)$$

If $Y_r \cap Y_s \neq \emptyset$ for $r \neq s \in \mathbb{Q}$, then $x_i + r = x_j + s$ for some representatives, so

$$x_i - x_j = s - r \in \mathbb{Q}, \quad (3.68)$$

forcing $x_i \sim x_j$; since A contains only one representative per class, $x_i = x_j$, hence $r = s$, a contradiction. So the sets $\{Y_r\}_{r \in \mathbb{Q} \cap [-1, 1]}$ are pairwise disjoint.

Since $A \subseteq [0, 1]$ and $r \in [-1, 1]$,

$$Y_r \subseteq [-1, 2] \quad \text{for every } r \in \mathbb{Q} \cap [-1, 1]. \quad (3.69)$$

For any $x \in [0, 1]$ there is a representative $x_i \sim x$ with $x - x_i \in \mathbb{Q} \cap [-1, 1]$, so $x \in Y_{x-x_i}$. Hence

$$[0, 1] \subseteq \bigcup_{r \in \mathbb{Q} \cap [-1, 1]} Y_r \subseteq [-1, 2]. \quad (3.70)$$

By translation invariance, $\mu(Y_r) = \mu(A)$ for every r . By monotonicity and countable additivity on the disjoint countable union,

$$\mu([0, 1]) \leq \mu\left(\bigcup_{r \in \mathbb{Q} \cap [-1, 1]} Y_r\right) = \sum_{r \in \mathbb{Q} \cap [-1, 1]} \mu(Y_r) = \sum_{r \in \mathbb{Q} \cap [-1, 1]} \mu(A) \leq \mu([-1, 2]). \quad (3.71)$$

Now $\mu([0, 1]) = 1$ and $\mu([-1, 2]) = 3$ (extending the usual length), so

$$1 \leq \sum_{r \in \mathbb{Q} \cap [-1, 1]} \mu(A) \leq 3. \quad (3.72)$$

But $\mathbb{Q} \cap [-1, 1]$ is countably infinite, so the middle sum is a countable sum of a single fixed value $\mu(A) \in [0, \infty]$:

- if $\mu(A) = 0$, the sum is 0, contradicting $1 \leq 0$;
- if $\mu(A) > 0$, the sum is $+\infty$, contradicting $\infty \leq 3$.

Either way we reach a contradiction. Hence no such measure μ can exist on all of $\mathcal{P}(\mathbb{R})$. ■

Remark 3.6. This is the price one pays for wanting a translation-invariant, monotone measure extending length: it cannot be defined on *every* subset of $\mathbb{R}$. This motivates restricting attention to a suitable σ -algebra (the Lebesgue measurable sets), which is the object to be constructed next.